

C PROGRAMMING : THREADING

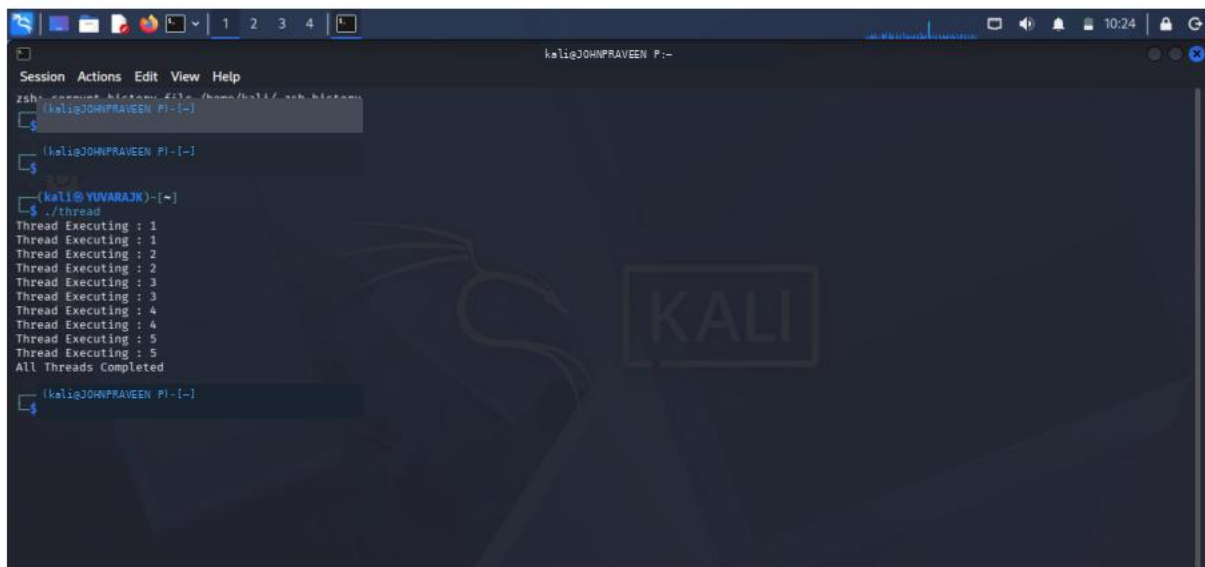


The screenshot shows a terminal window with the nano text editor open. The editor is editing a file named 'thread.c'. The code is as follows:

```
GNU nano 9.0 thread.c
#include <stdio.h>
#include <stdlib.h>
#include <pthread.h>
#include <unistd.h>
void *thread_function(void *arg)
{
    int i;
    for(i = 1; i <= 5; i++)
    {
        printf("Thread Executing : %d\n", i);
        sleep(1);
    }
    pthread_exit(NULL);
}
int main()
{
    pthread_t t1, t2;
    pthread_create(&t1, NULL, thread_function, NULL);
    pthread_create(&t2, NULL, thread_function, NULL);
    pthread_join(t1, NULL);
    pthread_join(t2, NULL);
    printf("All Threads Completed\n");
    return 0;
}
```

The nano editor's status bar at the bottom shows various keyboard shortcuts for editing and navigation.

OUTPUT :



The screenshot shows a terminal window with the output of the program. The user has run the command './thread' in a directory named 'thread'. The output is as follows:

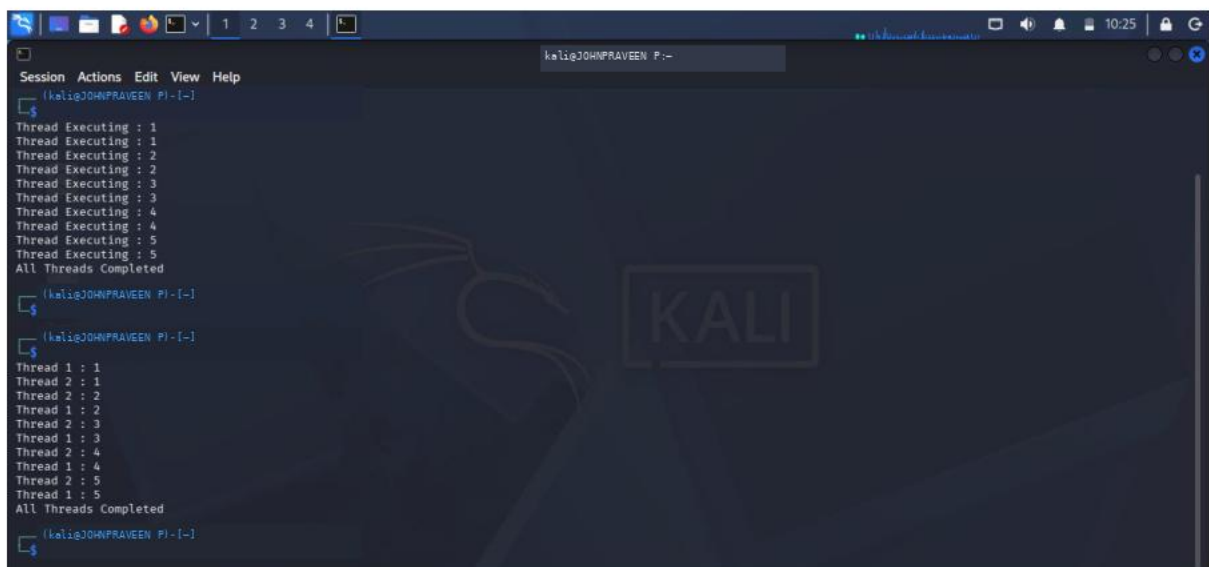
```
zsh: cannot history file (/home/kali/.zsh_history)
(kali@JOHNRAVEEN PI:~)
$
(kali@JOHNRAVEEN PI:~)
$
(kali@YUVARAJK)-[~]
$ ./thread
Thread Executing : 1
Thread Executing : 1
Thread Executing : 2
Thread Executing : 2
Thread Executing : 3
Thread Executing : 3
Thread Executing : 4
Thread Executing : 4
Thread Executing : 5
Thread Executing : 5
All Threads Completed
(kali@JOHNRAVEEN PI:~)
$
```

SHELL PROGRAMMING : THREADING



```
GNU nano 9.0 thread.sh
#!/bin/bash
task1()
{
  for i in 1 2 3 4 5
  do
    echo "Thread 1 : $i"
    sleep 1
  done
}
task2()
{
  for i in 1 2 3 4 5
  do
    echo "Thread 2 : $i"
    sleep 1
  done
}
task1 &
task2 &
wait
echo "All Threads Completed"
```

OUTPUT :



```
(kali@JOHNRAVEEN P:~)
$ ./thread.sh
Thread Executing : 1
Thread Executing : 1
Thread Executing : 2
Thread Executing : 2
Thread Executing : 3
Thread Executing : 3
Thread Executing : 4
Thread Executing : 4
Thread Executing : 5
Thread Executing : 5
All Threads Completed
(kali@JOHNRAVEEN P:~)
$
(kali@JOHNRAVEEN P:~)
$
Thread 1 : 1
Thread 2 : 1
Thread 2 : 2
Thread 1 : 2
Thread 2 : 3
Thread 1 : 3
Thread 2 : 4
Thread 1 : 4
Thread 2 : 5
Thread 1 : 5
All Threads Completed
(kali@JOHNRAVEEN P:~)
$
```